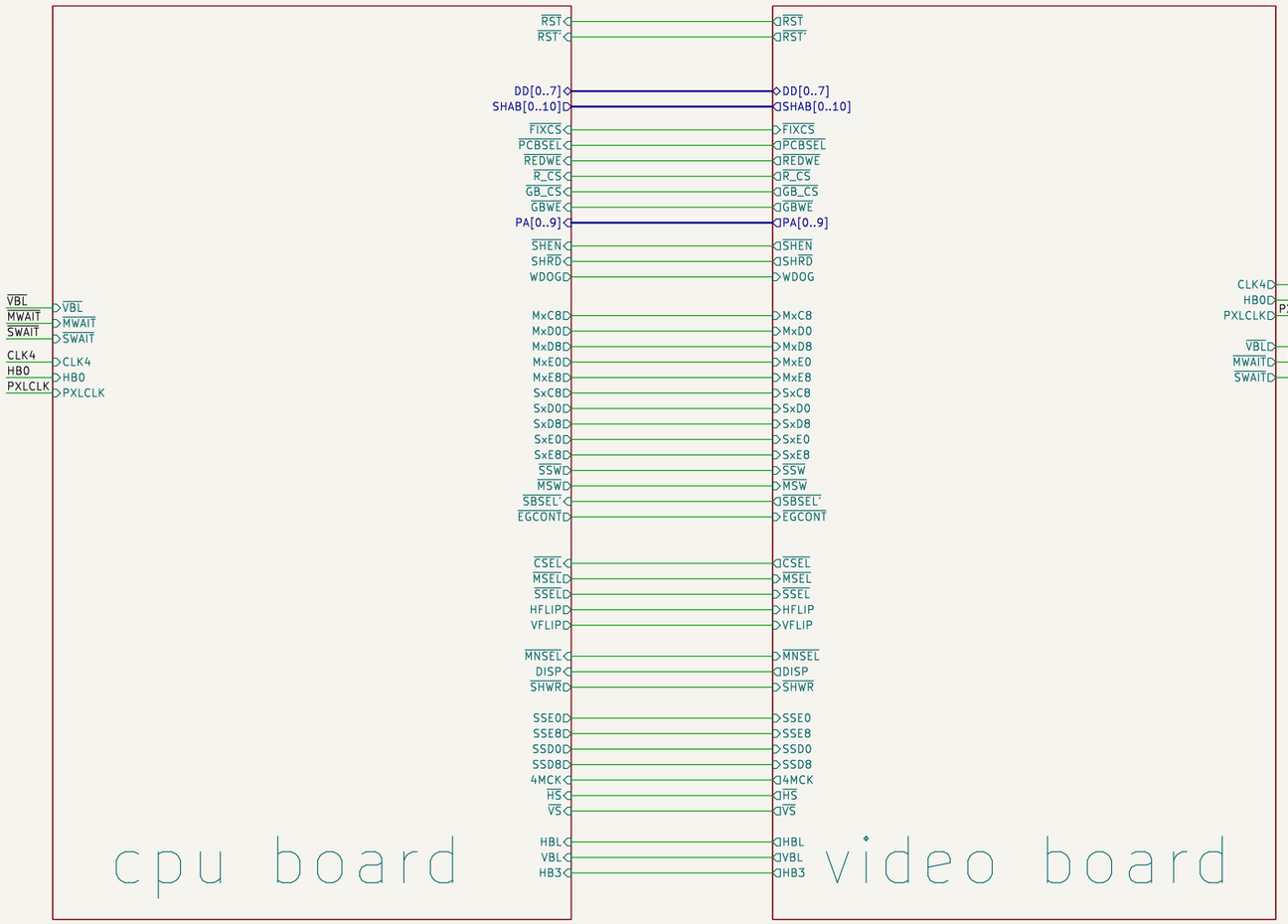


tehkanwch bootleg board

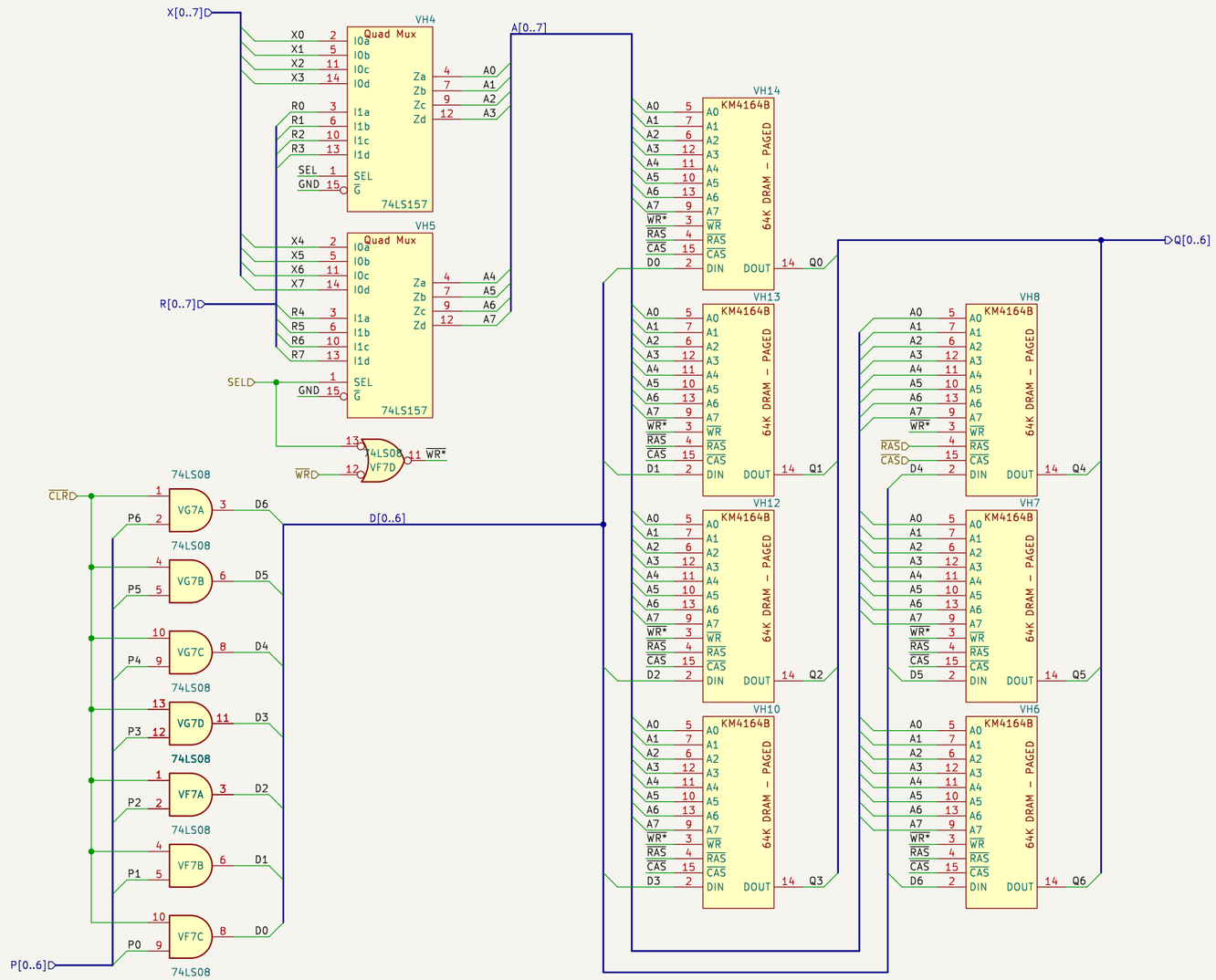
- original PCB has through-hole vias whereas the bootleg has filled vias
- original uses three custom packages from Mitsubishi with logic inside.
  These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
- debug edge connector on video board not present on bootleg
- cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
- different power amplifier for sound without heat sink on the bootleg
- slightly different frequencies for crystal oscillators (cost reduction?)
- device reference markers (silkscreen) on bootleg have reversed order on bootleg

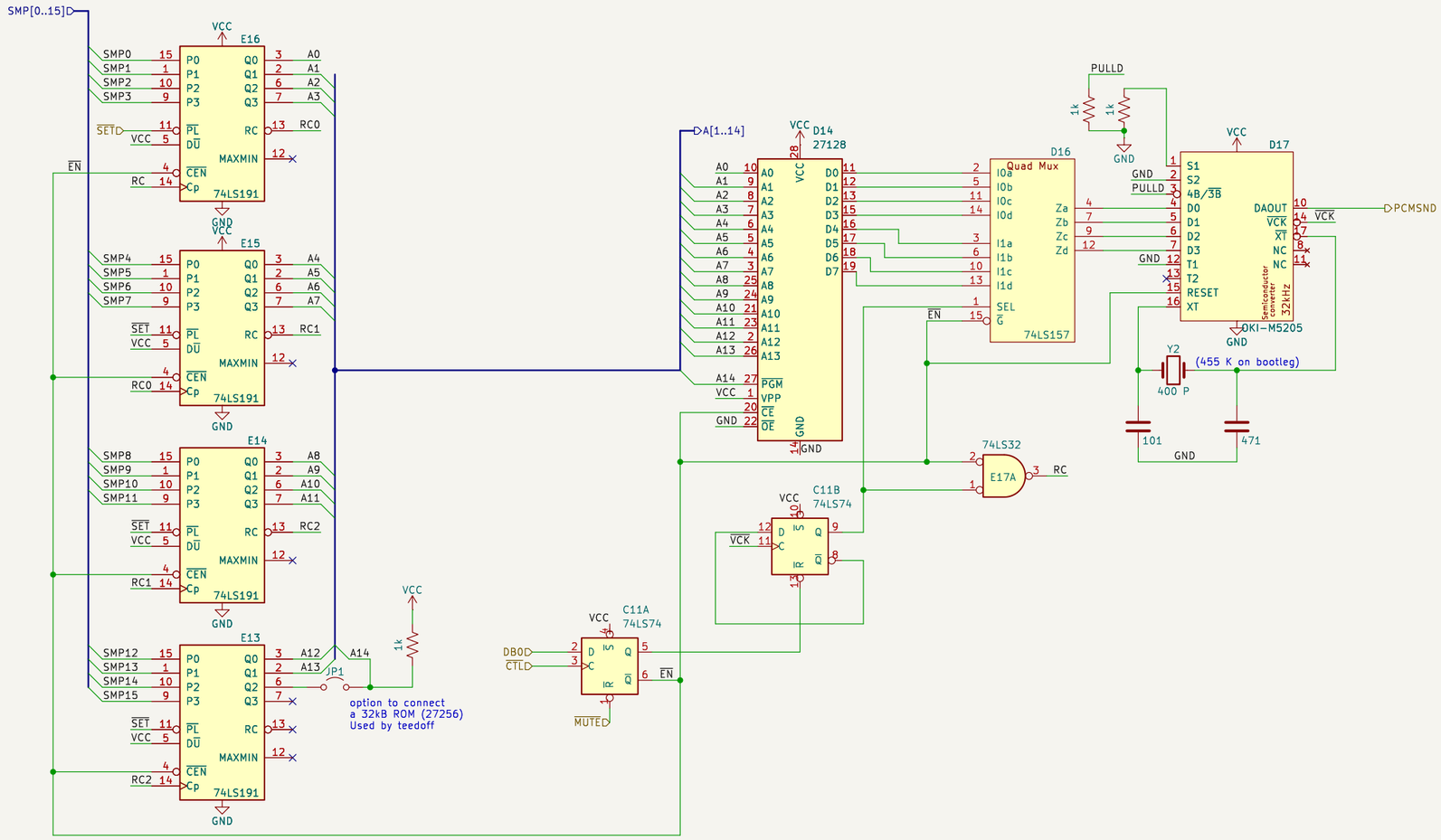
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Sheet: /
File: tehkanwch.kicad_sch

Title: TEHKAN WORLD CUP (tehkanwch bootleg)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/26


X[0..7]
R[0..7]
SELD
WRD
CLRD
P[0..6]
A[0..7]
D[0..6]
Q[0..6]
VH4
Quad Mux
74LS157
VH5
Quad Mux
74LS157
74LS08
VG7A
VG7B
VG7C
VG7D
VF7A
VF7B
VF7C
VF7D
VG7E
VF7E
VCC
GND
VH14
VH13
VH12
VH10
VH8
VH7
VH6
KM4164B
64K DRAM - PAGED
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Sheet: /video board/object/frame buffer B/
File: buffer.kicad_sch
Title: Line Buffer
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 8.0.5
Id: 2/26

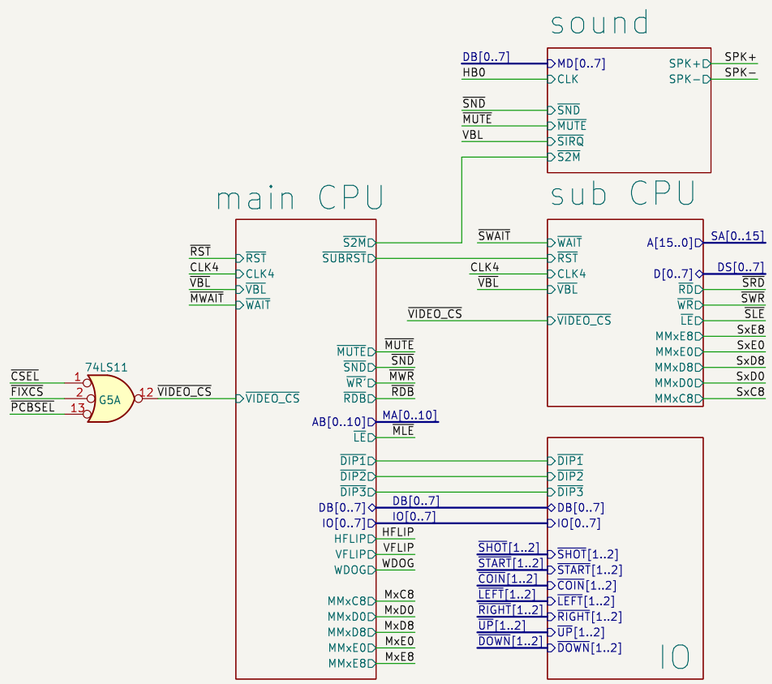


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Sheet: /cpu board/sound/PCM/
File: pcm.kicad_sch

Title: PCM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 5/26

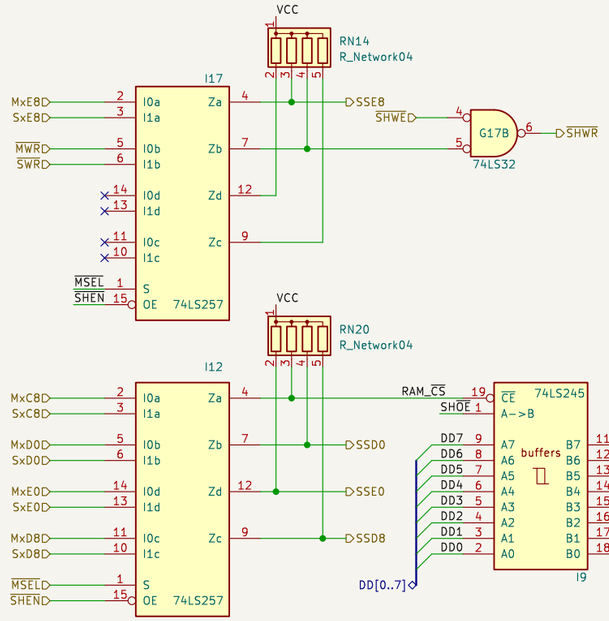


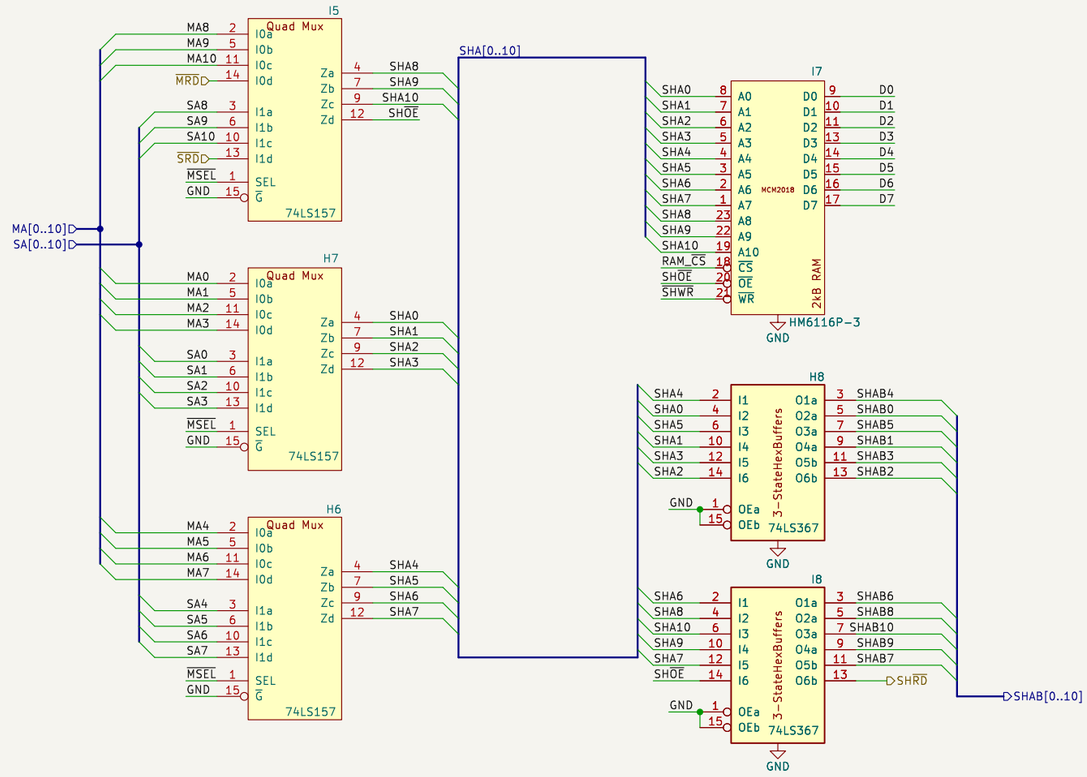
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Sheet: /cpu board/
File: cpu.kicad_sch

Title: CPU board (top board)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 9/26







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Sheet: /cpu board/shared/
File: shared.kicad_sch

Title: Shared Memory

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 10/26

timing

object

scroll

fix

color mixer

watchdog

address decoder

VJ1

connectors
to CPU board

VJ1 at PCB's top right
VJ2 at PCB's bottom right
pins counted from board bottom up,
pins at the top row of the connector
counted as even

VJ2

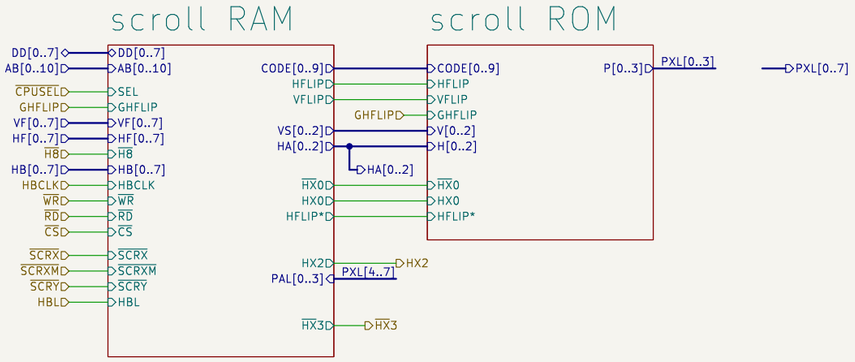
74LS08 VC6D

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Sheet: /video board/
File: video.kicad_sch

Title: Video board (bottom board)

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.5		Id: 11/26

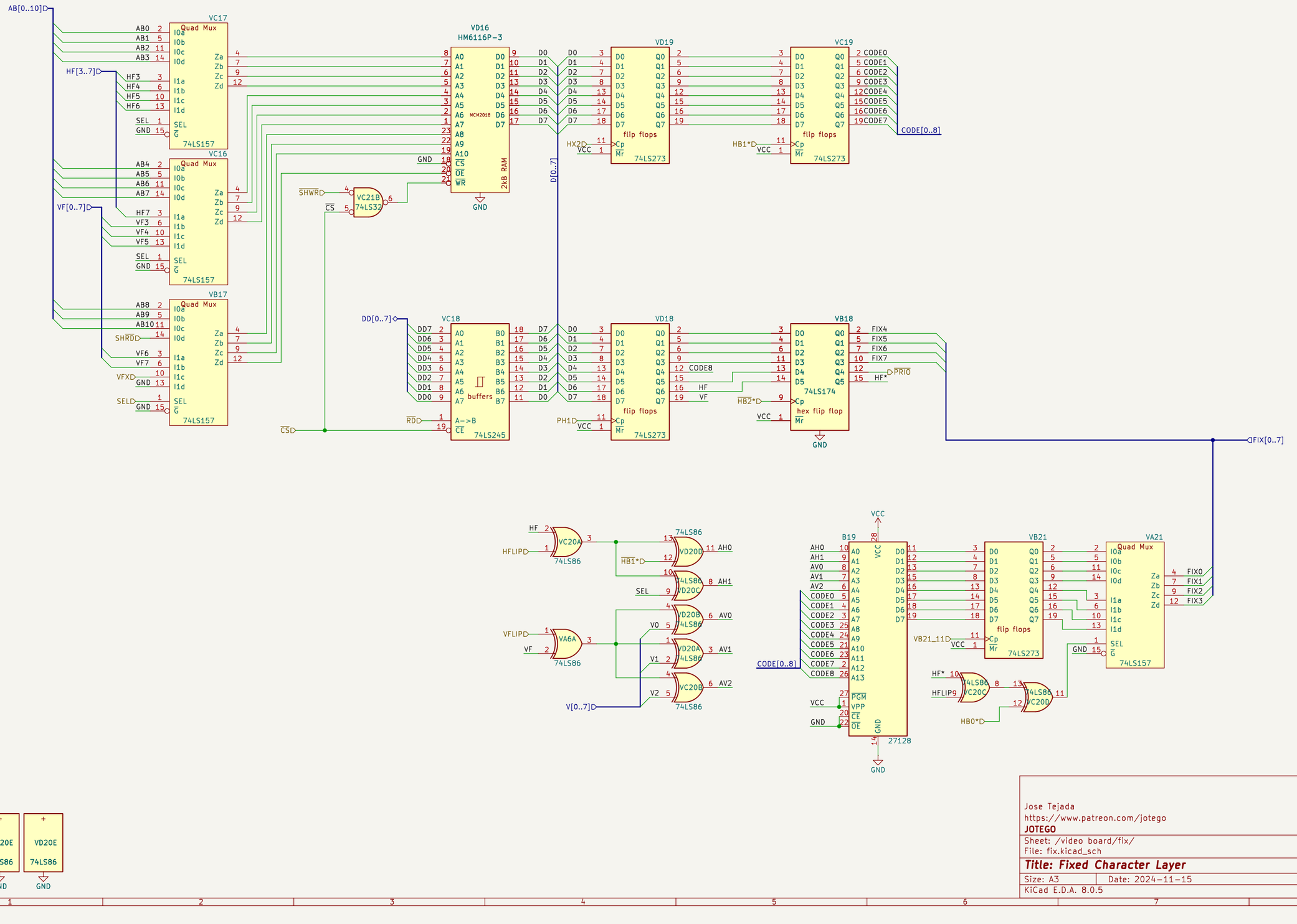


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Sheet: /video board/scroll/
File: scroll.kicad_sch

Title: Scroll Subsystem

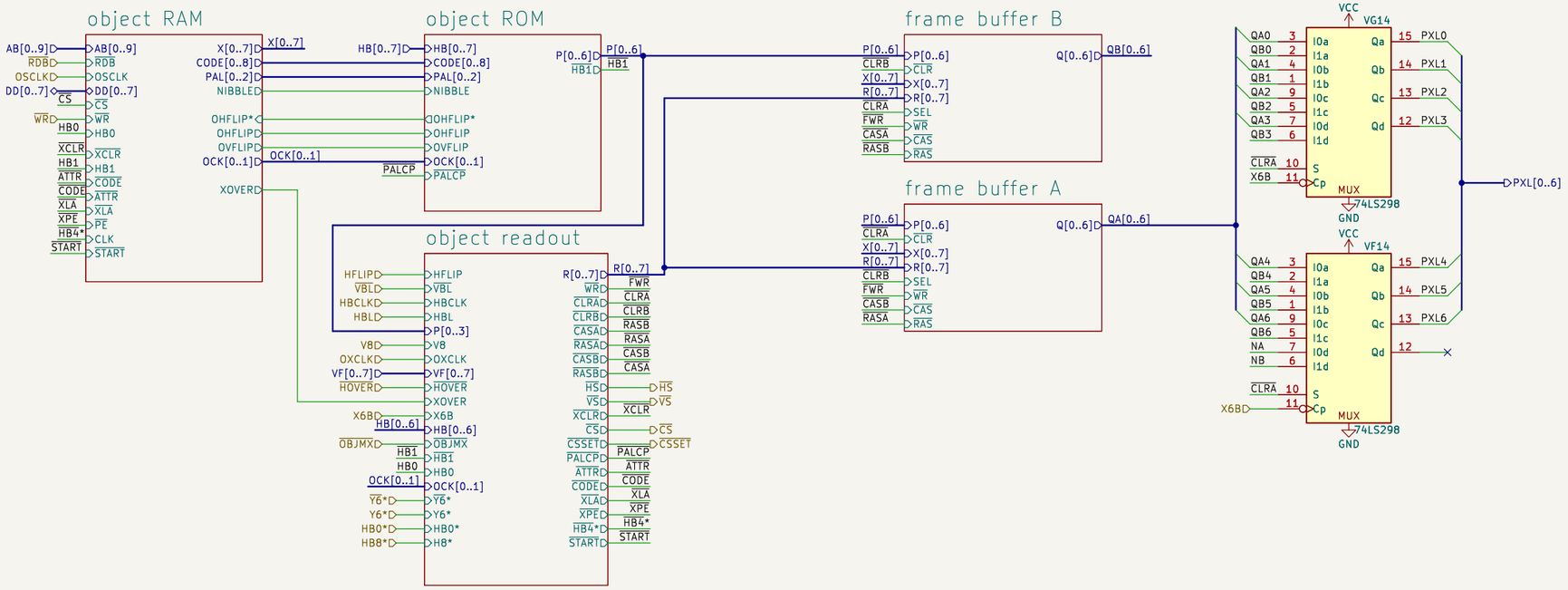
Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 12/26



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Sheet: /video board/fix/
File: fix.kicad_sch

Title: Fixed Character Layer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 13/26



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Sheet: /video board/object/
File: object.kicad_sch

Title: Object Subsystem

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 14/26

(3.57MHz on bootleg)

Y3
4.0MHz
101
101
GND
VI22A
4069
VI22B
4069
VH22A
74LS04
CLK4

clock divider by 3

VI20B
74LS107
VI20A
74LS107
Y6
X6
HCLK
6.144 MHz
VCC

K15
330
VJ15B
74LS04
C3
101
Y1
18.432
(18.000 on bootleg)
330
VJ15A
74LS04

VI16
74LS161
VI15
74LS161
V[0..8]
V0
VTC
VCLK
LSB
MSB
VG20F
74LS04

VI18
74LS161
VI19
74LS161
H0
3MHz
TCH
HCLK
VCC
GND
counter
VG19A
74LS74
VG20D
74LS04

VG21A
74LS74
VG19B
74LS74
HOVER
VCLK
H8
VG20C
74LS00
VH20B
H6
VG20B

74LS86
VH16D
VH16C
VH16B
VH16A
VF0
VF1
VF2
VF3
V1
V2
V3
V4
V5
V6
V7
VH15D
VH15C
VH15B
VH15A
VF4
VF5
VF6
VF7

VG18B
VFLIP
VF[0..7]

74LS20
VG17B
VG18C
VG17A
VH20D
74LS00
74LS08
74LS27
VG16A
VG20A
74LS04
VG16B
VE21B
X6B
TCH
HB6
HB7
HB5
VFX
HB1*
HB0*
HBCLK
object scan clock
OSCLK
HBL
OXCLK
SSDSET

VH17
3-StateHexBuffers
74LS367
HB[0..7]
VI17
3-StateHexBuffers
74LS367
X6
HCLK
Y6
H6
H7
Y6*
HB6
HB7
HBCLK

74LS86
VH18D
VH18C
VH18B
VH18A
HF0
HF1
HF2
HF3
74LS86
VH19A
VH19B
VH19C
VH19D
HF7
HF6
HF5
HF4
HF[0..7]
74LS08
VG18D
HFLIP

74LS32
VC22B
H8*
VD21A
74LS74
C22_13
101

VF21
74LS175
FOUR FLIP FLOP WITH DOUBLE RAIL OUTPUTS
HB0*
VB21_11
HB1*
HB1*
VFX
HB2*

VF20C
74LS08
VH20C
74LS00
VG21B
74LS74
VBL
VBL

VF22A
74LS74
PH1
VG22A
74LS74
VG22B
74LS74
VH22E
74LS04
VH22F
74LS04
VF20D
74LS08
DISP
H8*

VE20C
74LS32
PXLCLK

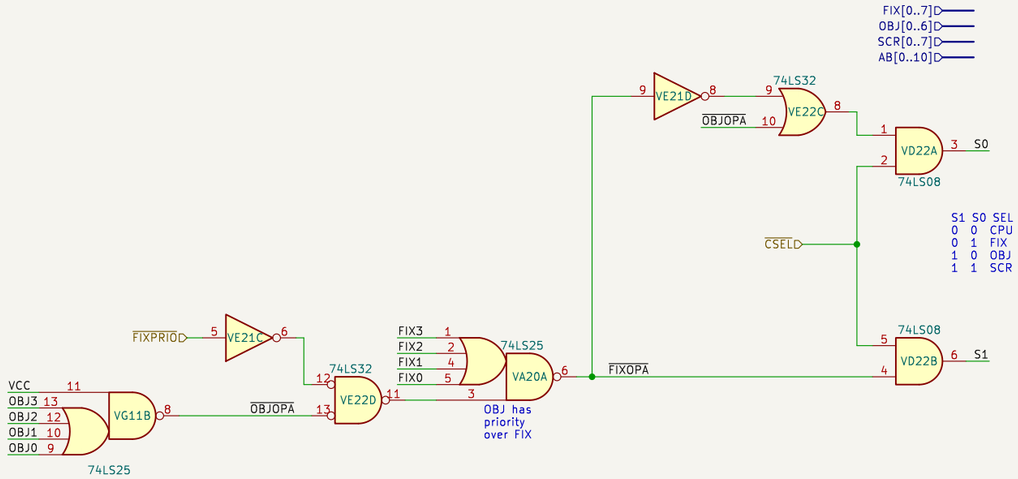
VE16C 74LS74
VG19C 74LS74
VJ15G 74LS04
VG18E 74LS08
VG15D 74LS10
VH20E 74LS00
VH18E 74LS86
VH19E 74LS86
VH16E 74LS86
VI20C 74LS107
VF22C 74LS74
VG22C 74LS74
VG20G 74LS04
VG21C 74LS74
VH22G 74LS04
VI22G 4069
VE21G 74LS04
VG17C 74LS20
VG16D 74LS27
VF20E 74LS08
VE20E 74LS32

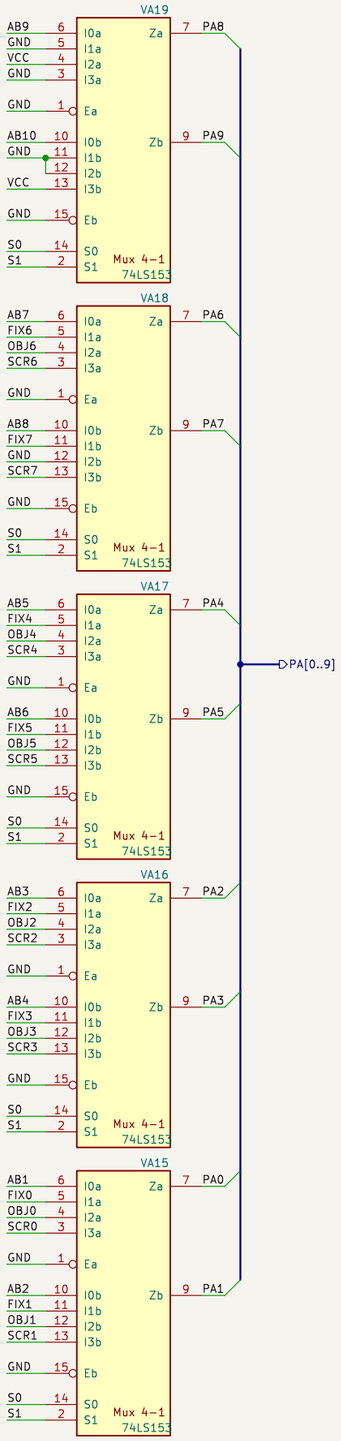
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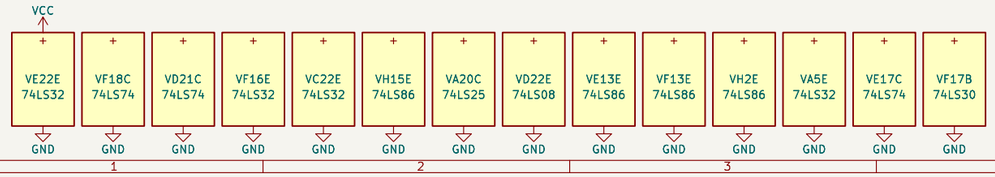
Sheet: /video board/timing/
File: timing.kicad_sch

Title: Video Timing

Size: A3 | Date: 2024-11-15 | Rev:
KiCad E.D.A. 8.0.5 | Id: 15/26





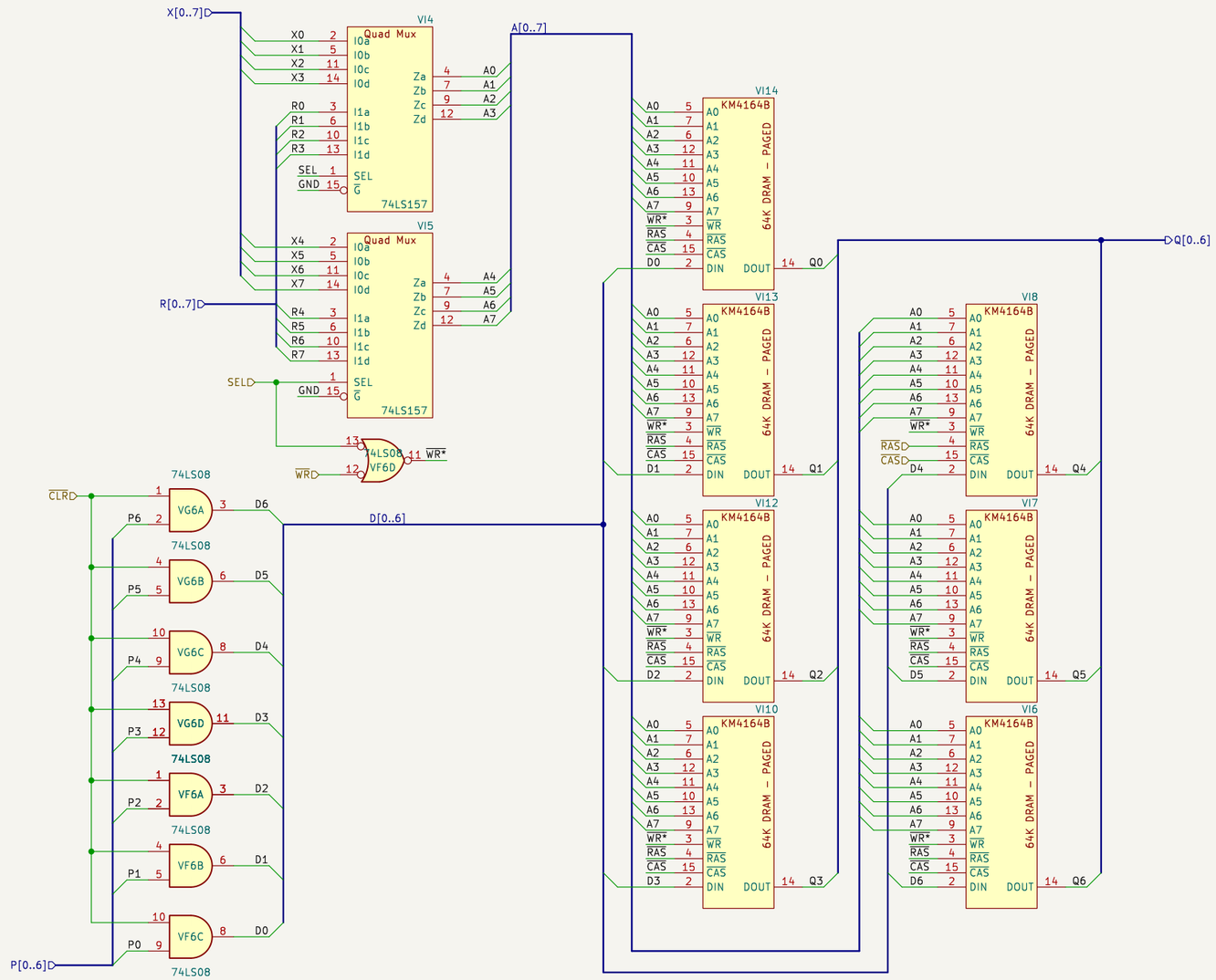


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Sheet: /video board/color mixer/
File: colmix.kicad_sch

Title: Color mixer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 16/26

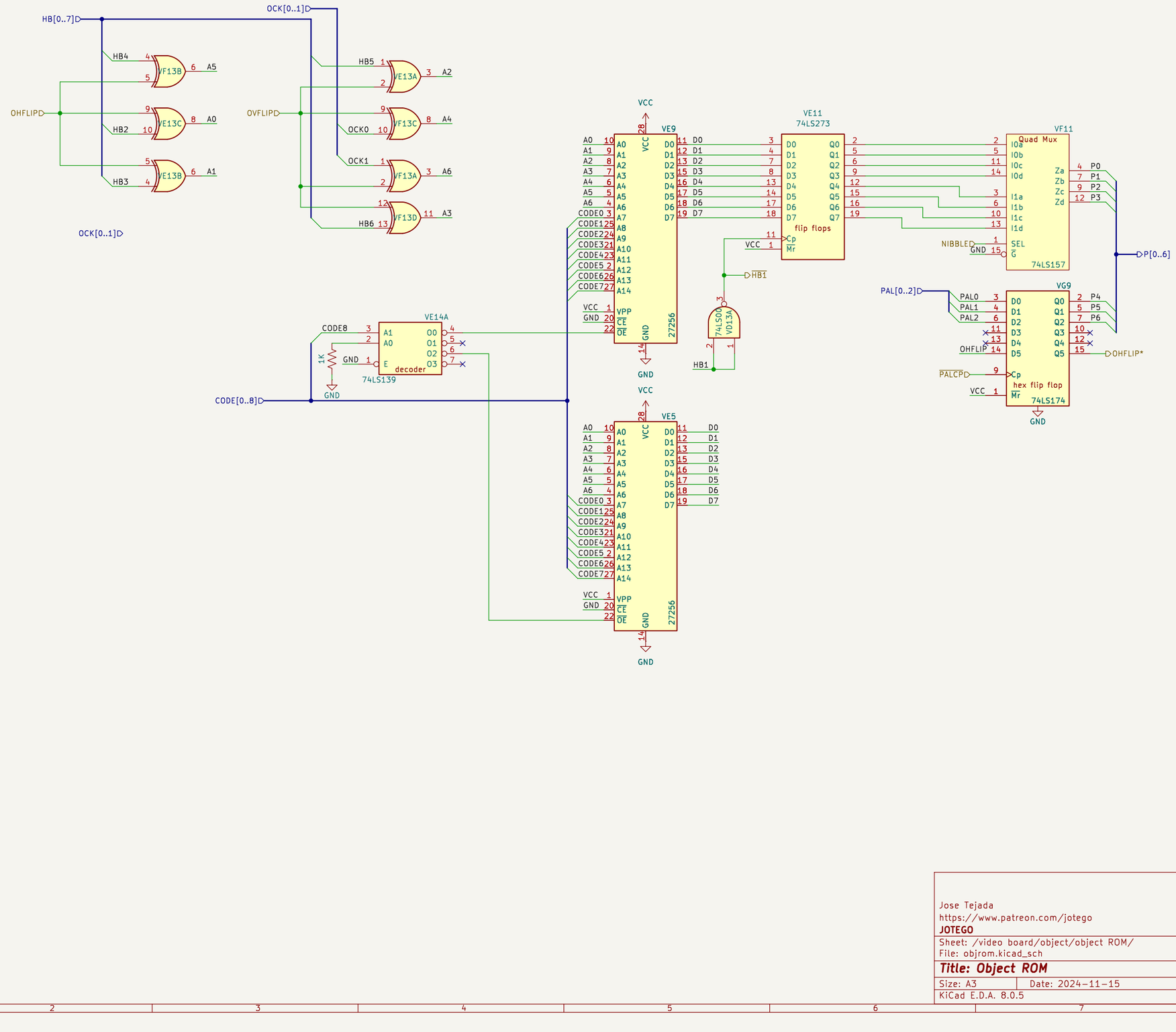


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Sheet: /video board/object/frame buffer A/
File: buffer.kicad_sch

Title: Line Buffer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 17/26

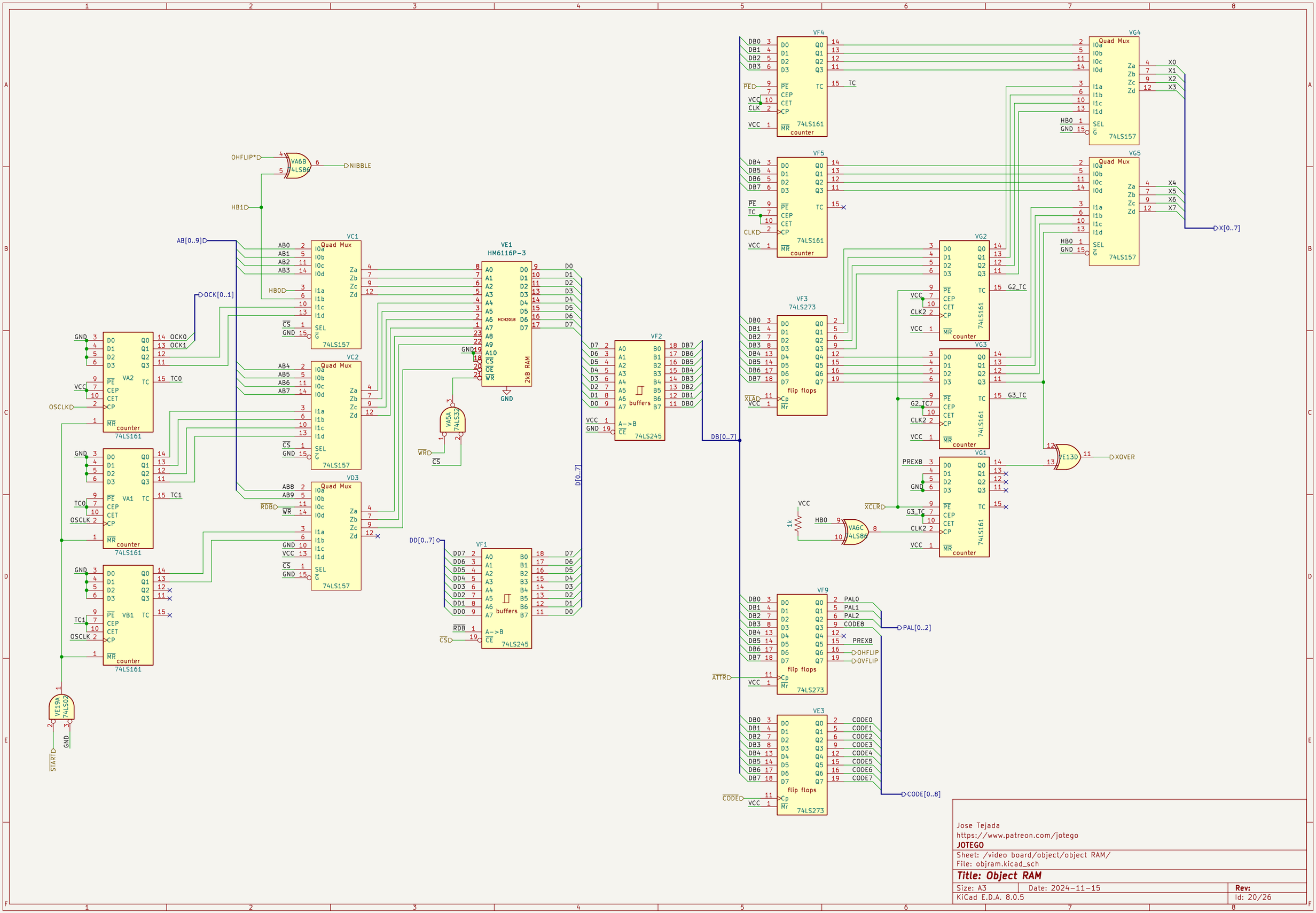


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Sheet: /video board/object/object ROM/
File: objrom.kicad_sch

Title: Object ROM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 19/26

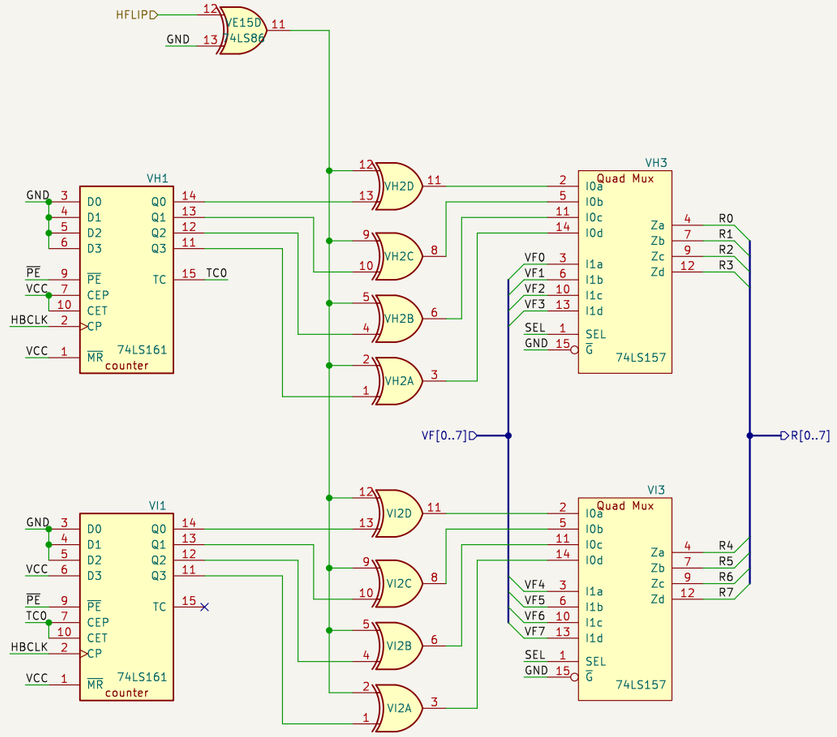


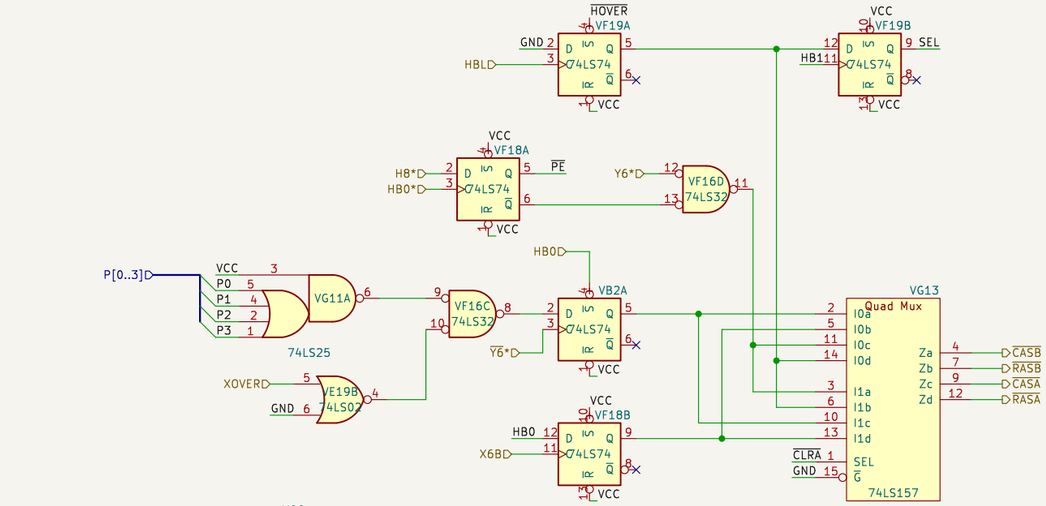
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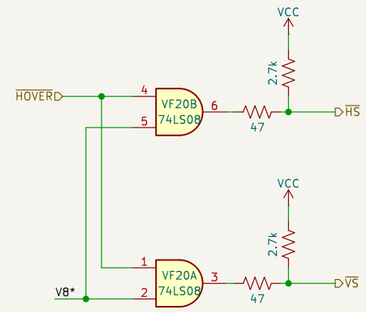
Sheet: /video board/object/object RAM/
File: objram.kicad_sch

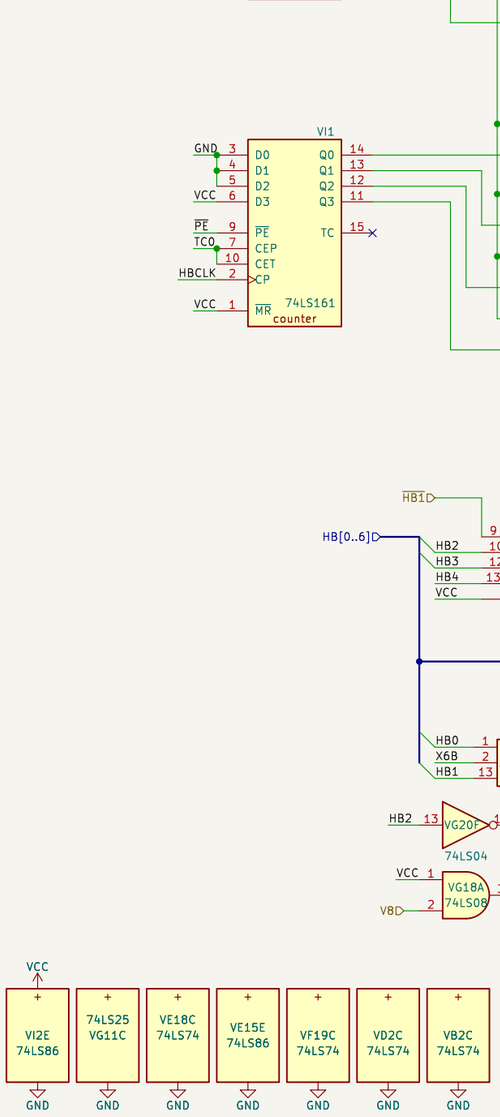
Title: Object RAM

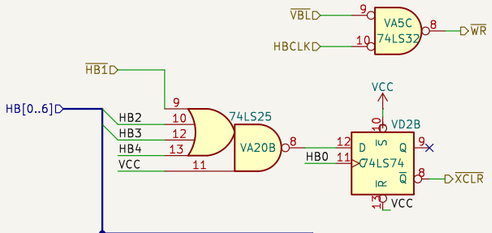
Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 20/26

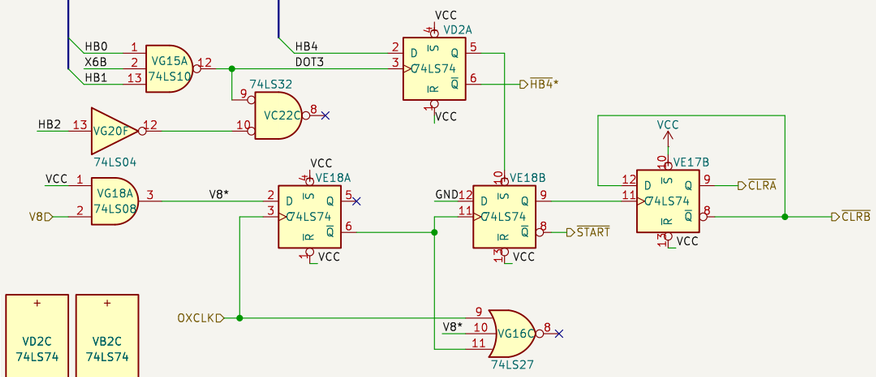


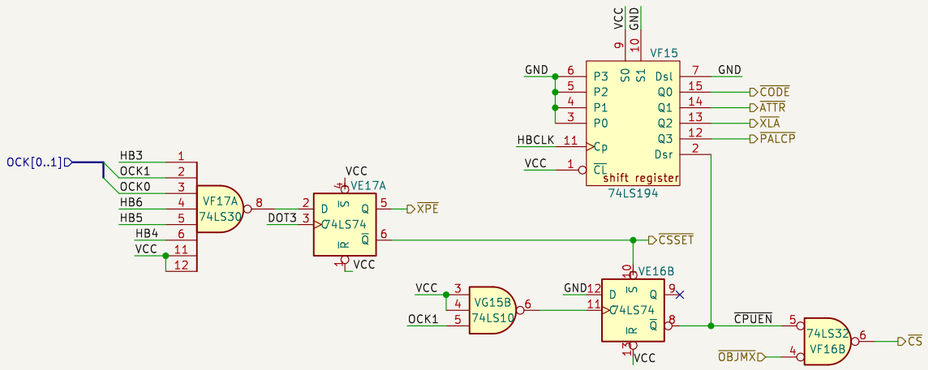










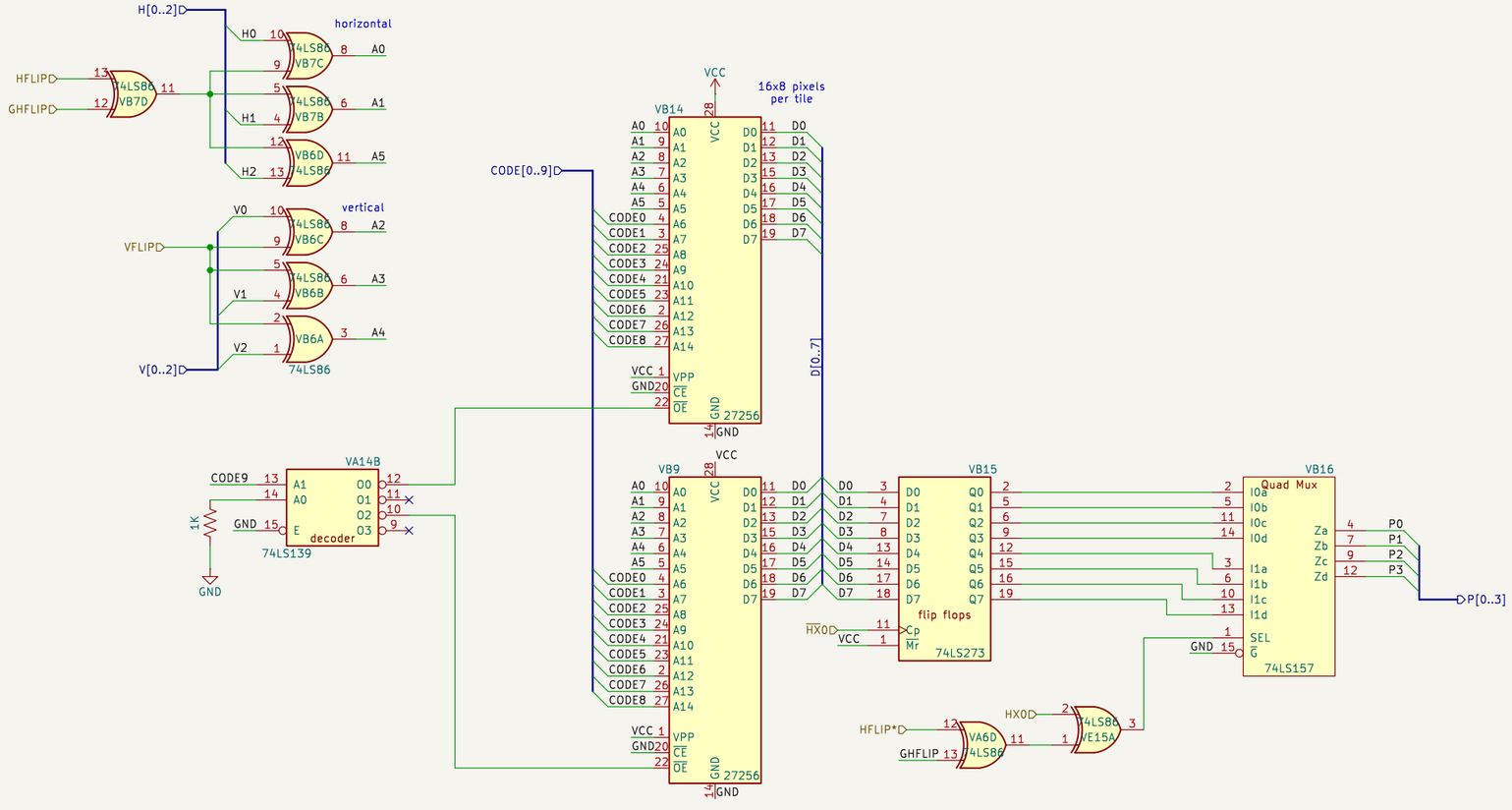


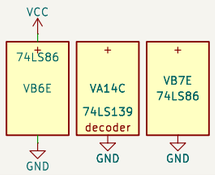
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Sheet: /video board/object/object readout/
File: objread.kicad_sch

Title: Object readout sequencer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 21/26



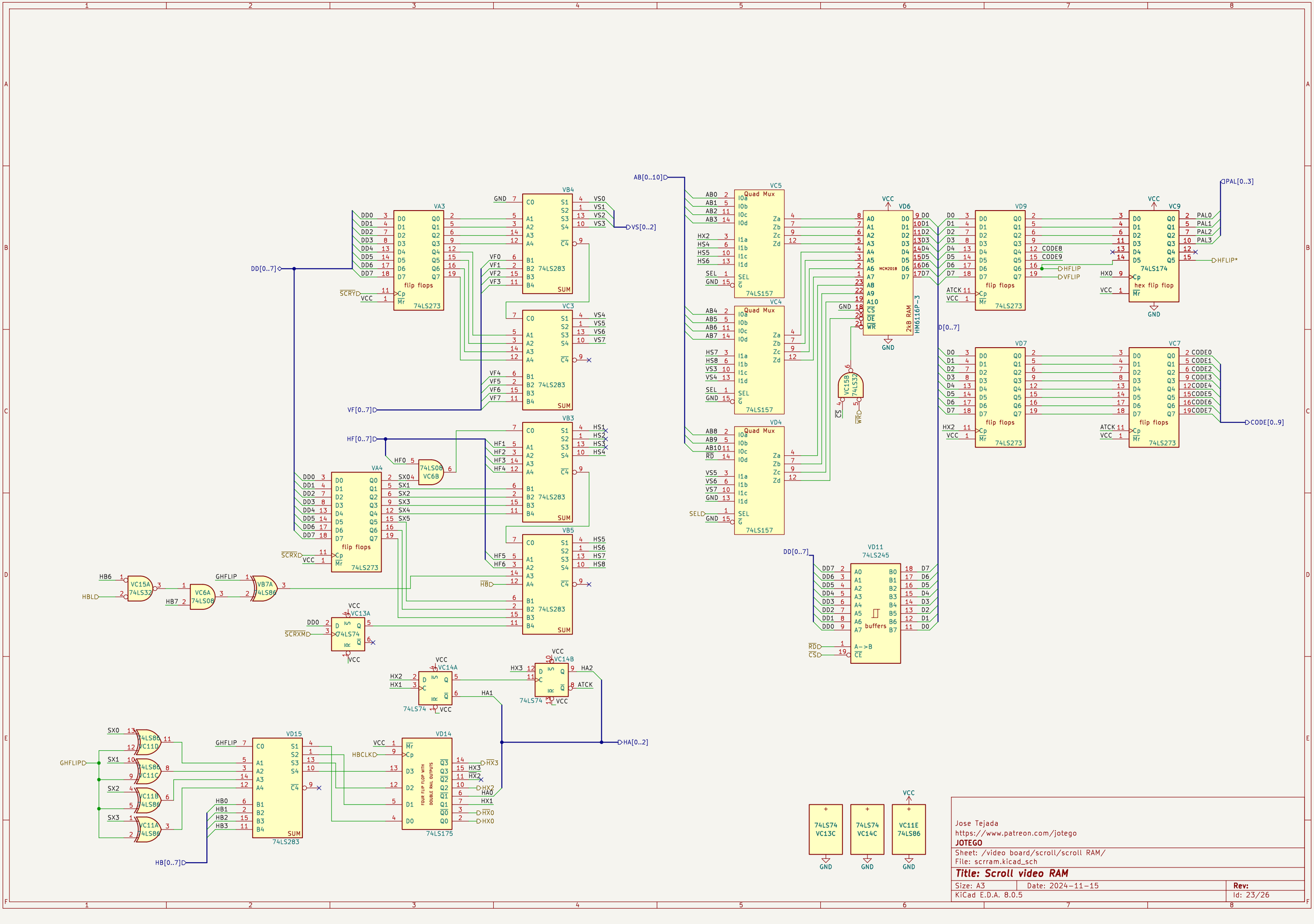


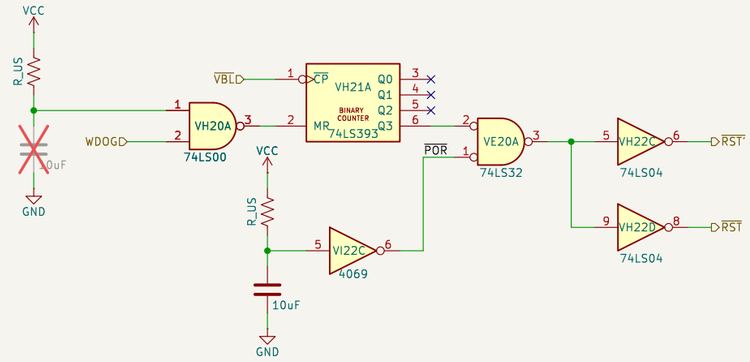
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Sheet: /video board/scroll/scroll ROM/
File: scrrom.kicad_sch

Title: scroll ROM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 22/26


AB[0..10]
DD[0..7]
VF[0..7]
HF[0..7]
HB[0..7]
VS[0..2]
HA[0..2]
PAL[0..3]
CODE[0..9]
HFLIP*
VA3
flip flops
74LS273
VB4
VB5
VB3
74LS283
SUM
VA4
VC3
VC4
VC5
Quad Mux
74LS157
VD4
VD6
MCM2018
2kB RAM
HM6116P-3
VD9
VD7
VC7
74LS174
hex flip flop
VC9
VD11
74LS245
buffers
VD14
74LS175
VD15
SCRY
SCRX
SCRXM
HBLD
HBCLK
GHFLIP
HFLIP
VFLIP
ATCK
HX0
HX1
HX2
HX3
HA0
HA1
HA2
HX[0..3]
HBX[0..3]
SX0
SX1
SX2
SX3
SX4
SX5
VC15A
74LS32
VC6A
74LS08
VB7A
74LS86
VC13A
74LS74
VC14A
VC14B
VC15B
VC11D
VC11C
VC11B
VC11A
VC6B
VC13C
VC14C
VC11E
GND
VCC
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JOTEGO
Sheet: /video board/scroll/scroll RAM/
File: scrram.kicad_sch
Title: Scroll video RAM
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 8.0.5
Id: 23/26



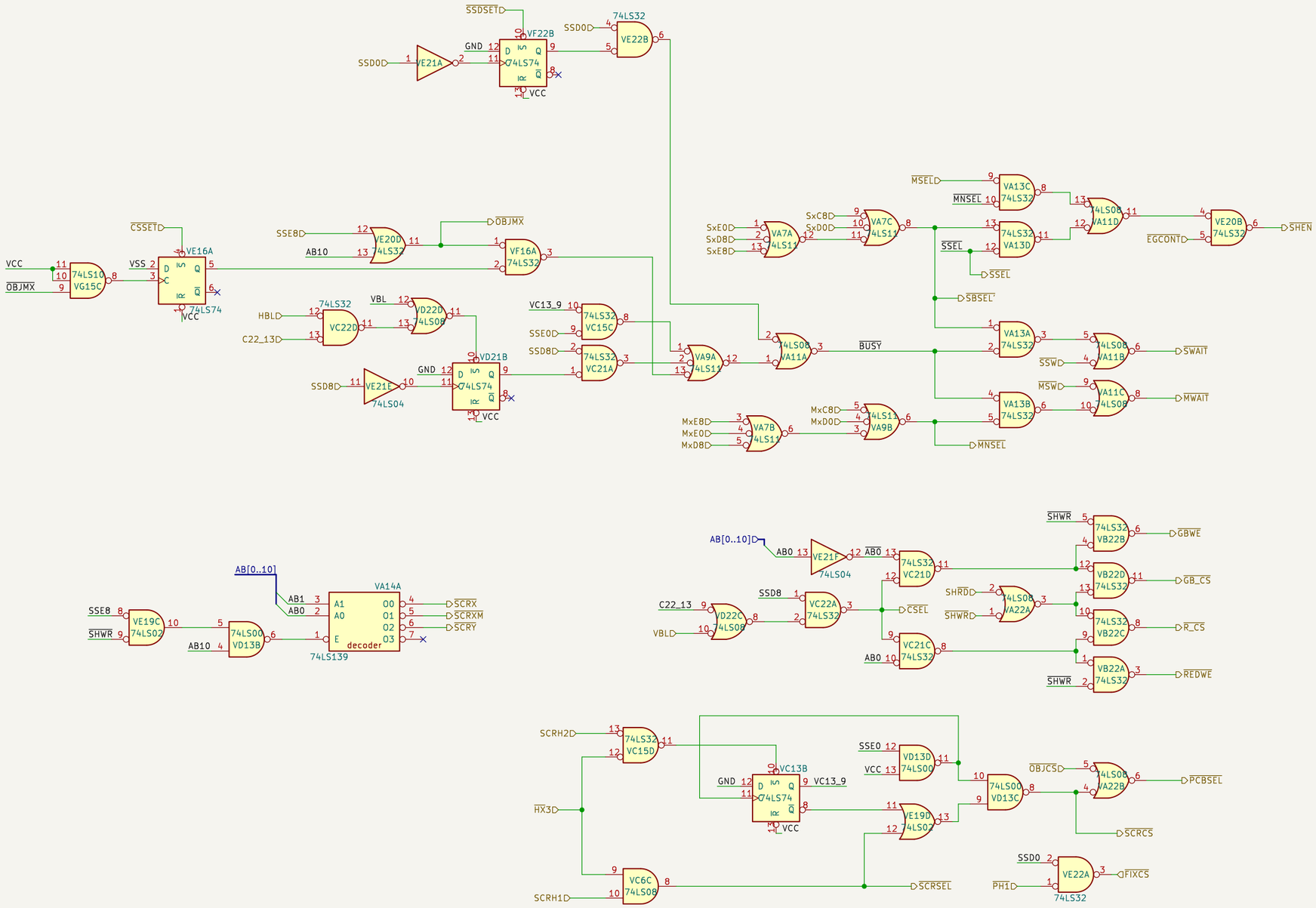


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Sheet: /video board/watchdog/
File: watchdog.kicad_sch

Title: Watchdog Circuit

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 24/26

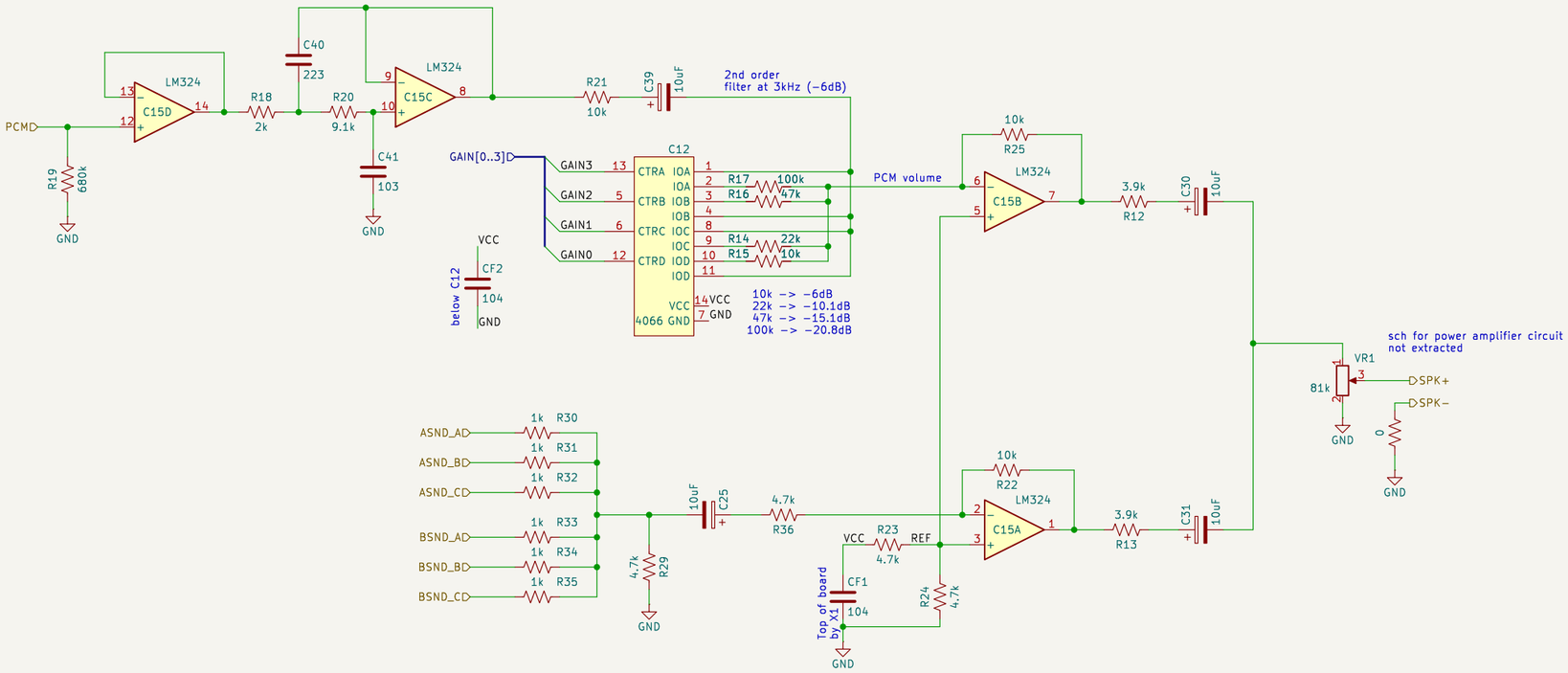


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Sheet: /video board/address decoder/
File: decoder.kicad_sch

Title: Address decoder

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 25/26



Passive component references
when shown, came from original PCB.
The bootleg did not have
references for passives.

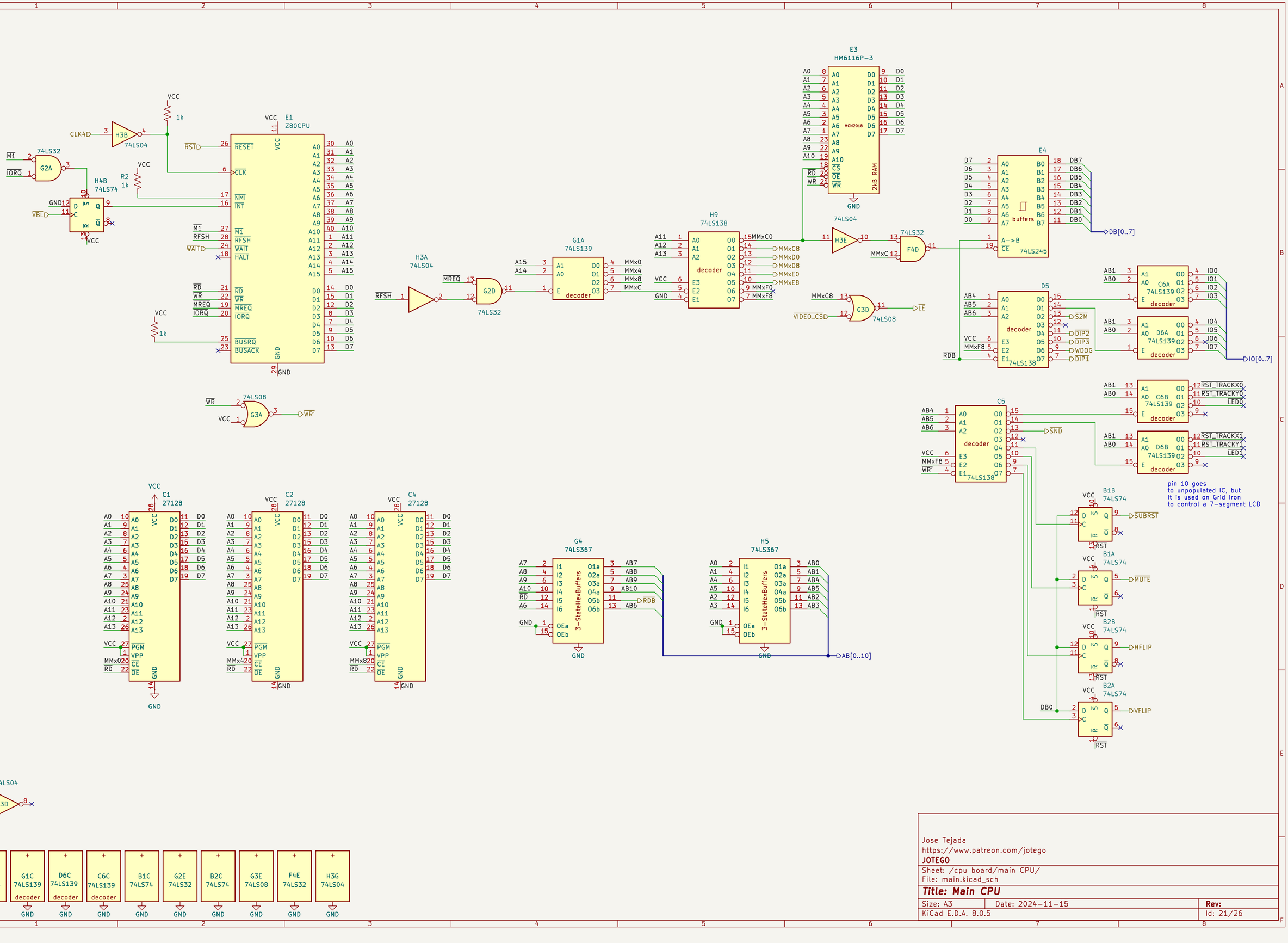
Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

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Sheet: /cpu board/sound/analog/
File: analog.kicad_sch

Title: Sound amplifiers

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 26/26



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Sheet: /cpu board/main CPU/
File: main.kicad_sch

Title: Main CPU

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 21/26

Taps to unpopulated sockets

NMICLR

74LS08 E2C

G6B 74LS74

GND SND VCC

M1 IORQ 74LS32 G17C

I16A 74LS74

GND SIRQ VCC

H17A 74LS04

MREQ RFSH 74LS32 G17A

H16B 74LS139 decoder

A15 A14 O0 O1 O2 O3 MMx0 MMx4 MMx8 MMx0C

H16A 74LS139 decoder

E17B 74LS32

MMx8 WRB AB1 AB0 O0 O1 O2 O3 PCMSET PCMCTL GAINSET NMICLR

VCC G13 Z80CPU

MUTE CPUCLK CLK RESET NMI INT M1 RFSH WAIT HALT RD WR MREQ IORQ BUSRQ BUSACK A0–A15 D0–D7 GND

1K 1K VCC

H13 27128

A0–A14 D0–D7 VCC VPP PGM CE OE GND

I14 HM6116P-3

A0–A10 D0–D7 MMx8 CS RDB WRB OE WR 2kB RAM

D[0..7]

F12 YM2149

D0–D7 DA0–DA7 A8 A9 BDIR BC1 BC2 CLOCK SEL RESET TEST1 VCC VSS GND

ASND_A ASND_B ASND_C IOA0–IOA7 IOB0–IOB7 CNT0–CNT13 1k VCC

CNT[0..13]

F15 YM2149

BSND_A BSND_B BSND_C SMP0–SMP15

SMP[0..15]

74LS04 H17C CLK CPUCLK

I16B 74LS74 PSGCLK

MD[0..7]

E11 E12 3-state flip flops 74LS374

DB0–DB7 MD0–MD7

F17 74LS367 3-StateHexBuffers

RD A0 A1 WR IORQ RDB ABO AB1 IORQB OEa OEb

F4A 74LS32 MMxC SND

G17D 74LS32 WRB MMxC S2M

F11 74LS245 buffers

DB0–DB7 RDB A–>B CE

E10D 74LS08 MMx8 MMxC

D12 74LS174 hex flip flop

DB1 DB2 DB3 DB0 GAINSEL MUTE Cp Mr Q0–Q5 GAIN1 GAIN2 GAIN3 GAIN0

GAIN[0..3]

PCM

PCMCTL PCMSET DBO MUTE SMP[0..15] CNT[0..13] CTL SET DBO MUTE SMP[0..15] DA[1..14] PCMSND

analog

ASND_A ASND_B ASND_C BSND_A BSND_B BSND_C GAIN[0..3] PCMSND ASND_A ASND_B ASND_C BSND_A BSND_B BSND_C GAIN[0..3] PCM SPK+ SPK− SPK+ SPK−

File: analog.kicad_sch

74LS02 F16B F16D F16A G16B G16D G16A

ABDIR ABC2 ABC1 BBDIR BBC2 BBC1

IORQB AB1 H17F 74LS04 E17D E17C 74LS32 IOA1

BDIR	BC2	BC1	
0	X	0	inactive
1	0	0	latch address
1	1	0	data write
0	0	1	latch address
0	1	1	data read

G6A 74LS74 74LS02 G16C

H17G 74LS04 GND

H16C 74LS139 decoder GND

E10E 74LS08 GND

I16C 74LS74 GND

F16E 74LS02 GND

E2E 74LS08 GND

G16E 74LS02 GND

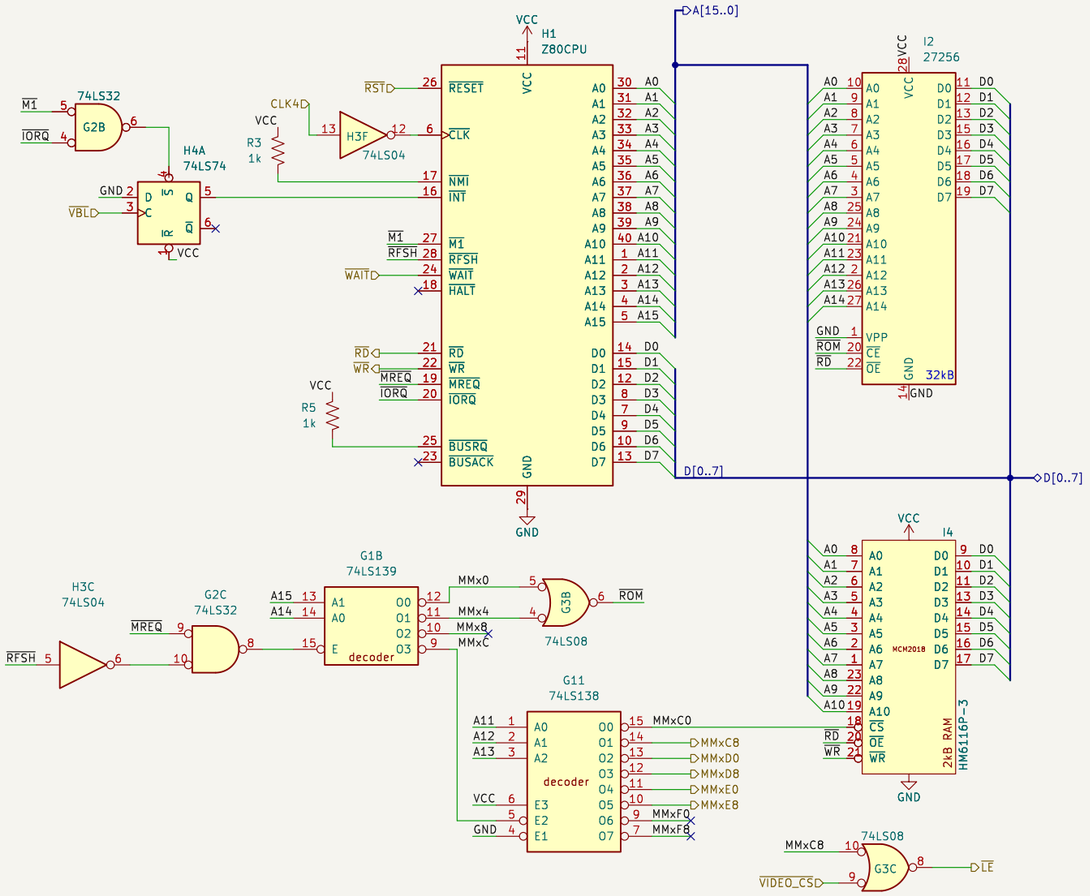
G6C 74LS74 GND

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Sheet: /cpu board/sound/
File: sound.kicad_sch

Title: Sound

Size: A3	Date: 2024-11-15	Rev:
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Sheet: /cpu board/sub CPU/
File: subcpu.kicad_sch

Title: Subordinate CPU

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 23/26

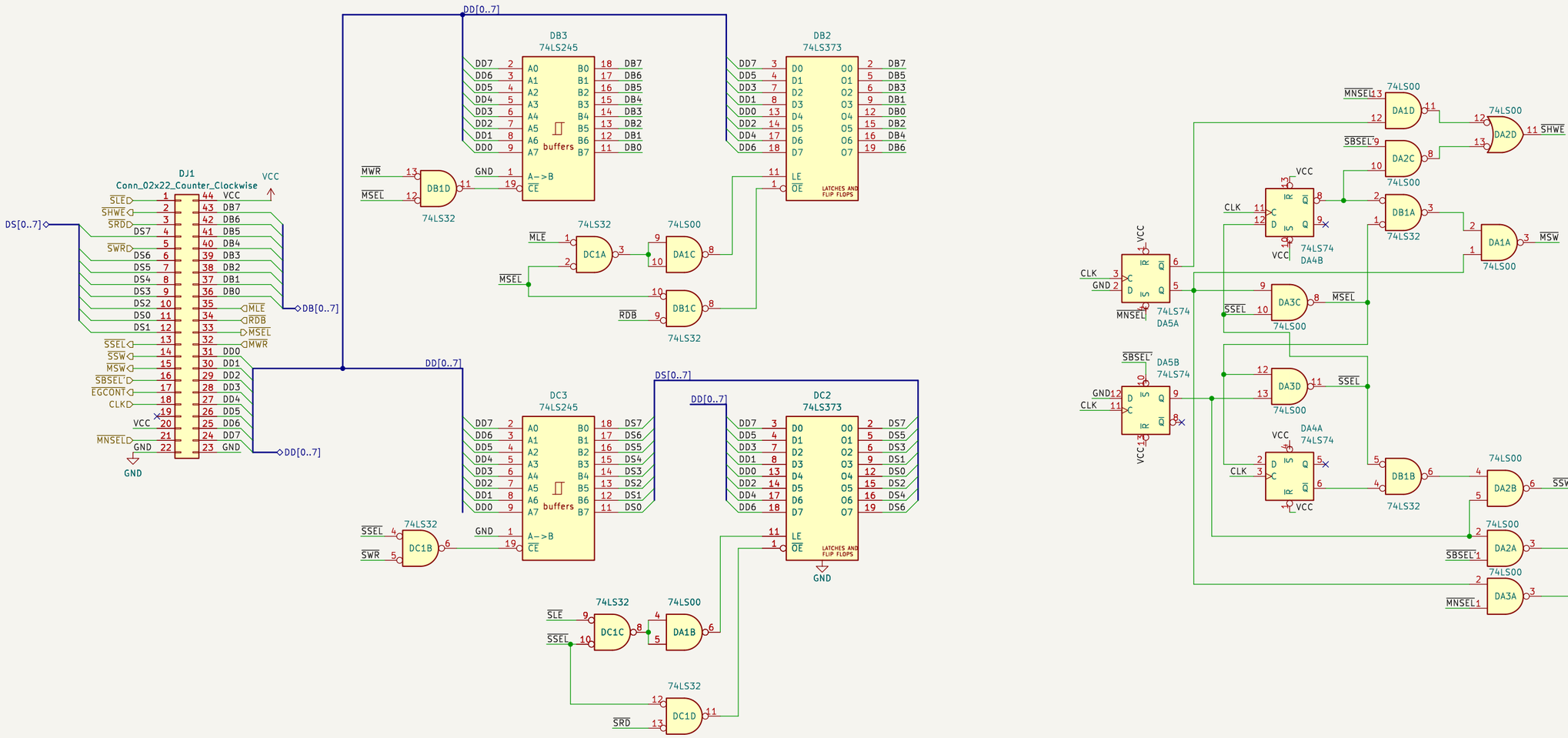
F5
SW_DIP_x08
E6
74LS244
GND
E5
74LS244
SW_DIP_x08
F6
DIP2
DIP3
B3
3-StateHexBuffers
74LS367
A2
SW_DIP_x04
DIP1
DB[0..7]
VCC
RN18
R_Network08_US
PU0
PU1
PU2
PU3
PU4
PU5
PU6
PU7
DOWN[1..2]
UP[1..2]
RIGHT[1..2]
LEFT[1..2]
maybe through some pullup?
LEFT1
ROD1
B5E
74LS04
RIGHT1
ROC1
UP1
ROF1
B5D
DOWN1
ROE1
LEFT2
ROB1
B7B
RIGHT2
ROA1
UP2
ROH1
B7C
DOWN2
ROG1
C10
C9
C8
C7
Quad Mux
74LS157
D10
D9
D8
D7
IO0
IO1
IO4
IO5
IO[0..7]
COIN[1..2]
COIN1
COIN2
START1
START2
START[1..2]
RN17
R_Network04
B12
IO2
C
B11
IO3
B4
IO7
SHOT1
SHOT2
100RD2
100RD1
SHOT[1..2]
IO side of the board contains unpopulated footprints. The schematics do not reflect the full PCB on this side.
RN13
RN19
Pull-up pin order may not match the PCB
RPU8
RPU9
RPU10
RPU11
RPU12
RPU13
RPU14
RPU15
RPU16
RPU17
RPU18
RPU19
RPU20
RPU21
RPU22
RPU23
220
B5G
74LS04
B7G
74LS04

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Sheet: /cpu board/IO/
File: io.kicad_sch

Title: Cabinet Input/Output

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 24/26



This circuit takes three small boards on the original PCB.
These boards were manufactured by Mitsubishi and contain the circuits seen on the bootleg
– MA7053 x2, 28 pin each
– MA7054 x1, 15 pins
The layout around the daughter board on the bootleg differs slightly from the original in order to adjust the pinout to a single 44–pin board.

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Sheet: /cpu board/daughter/
File: daughter.kicad_sch

Title: Small daughter board

Size: A3	Date: 2024–11–15	**Rev:**
KiCad E.D.A. 8.0.5		Id: 25/26

Resistor networks reference:
RN1 = E9 top
RN2 = below RN1
...
RN11 = above RN12
RN12 = E7 bottom

R from pin 1 to 2 = 2.2kOhm

SYMBOL DOES
NOT MATCH
INTERNAL NETWORK

G and R
are likely
swapped

MSB to common pin 222 Ohm
Each bit up gets 2x (verified)

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Sheet: /cpu board/palette/
File: palette.kicad_sch

Title: Color Palette

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.5		Id: #/26